

Hindi → Marathi Neural Machine Translation

IndicSeq2Seq: Bidirectional LSTM Seq2Seq with Bahdanau Attention

1. Abstract

This report presents **IndicSeq2Seq**, a Neural Machine Translation (NMT) system for Hindi-to-Marathi translation. The model uses a 2-layer Bidirectional LSTM Encoder with Bahdanau Attention and a 2-layer Unidirectional LSTM Decoder, trained on a parallel Hindi-Marathi corpus with a joint SentencePiece tokenizer (32,000 vocab).

Training was done on Kaggle T4 x2 GPUs over 16 epochs. The final model achieves:

Metric	Value
Validation Loss	6.487
BLEU (val set)	7.97
CHRF (val set)	32.41
Best Decoding BLEU (qualitative set)	5.69 (Beam + Repetition Blocking)
Best Decoding CHRF (qualitative set)	45.27 (Beam Search)

2. Introduction

Hindi and Marathi are both Indo-Aryan languages sharing the Devanagari script and a significant vocabulary overlap. Despite this similarity, Hindi-Marathi MT is a low-resource problem: parallel corpora are small and noisy, both languages have complex morphology, and named entities are undertrained.

Project Goals:

- 1. Build a classical Seq2Seq NMT system with attention for Hindi-Marathi.
- 2. Train it under compute-constrained conditions (Kaggle free tier).

3. Improve inference quality through principled decoding strategies.
4. Analyse failure modes and understand model limitations.

3. Dataset and Preprocessing

3.1 Corpus

Split	Sentences	Approx. Tokens (Hindi)
Train	~215,000	~3.5M
Validation	~10,000	~160K

Source: Public Indic NLP parallel corpora covering news, literature, and government domains.

3.2 Preprocessing Steps

- Remove empty lines and whitespace-only pairs.
- Filter pairs with source-to-target length ratio > 3.0 .
- Remove duplicate sentence pairs.
- Strip web-scraping artifacts and excessive punctuation.

3.3 Tokenizer

A **shared SentencePiece BPE** tokenizer is trained jointly on both Hindi and Marathi text. Using a shared vocabulary allows the model to leverage cognate subwords (words with identical script in both languages), reduces out-of-vocabulary rates, and enables shared encoder-decoder embeddings.

Vocabulary size: 32,000 tokens.

Special tokens: PAD (0), UNK (1), SOS (2), EOS (3).

4. Model Architecture

4.1 Overview

The model follows an Encoder-Decoder architecture with attention:

```

Hindi Input Tokens
|
[Embedding Layer: 32000 x 256]
|
[BiLSTM Encoder: 2 layers, hidden=512 per direction]
| --> Encoder outputs: dim=1024 (forward + backward concat)
| --> Hidden state: projected to dim=512
|
[Bahdanau Attention]
| e(t,i) = v . tanh(W1.h_i + W2.s_{t-1})
| alpha(t,i) = softmax(e(t,i))
| c_t = sum over alpha(t,i) * h_i
|
[Unidirectional LSTM Decoder: 2 layers, hidden=512]
| Input per step: [embed(y_{t-1}); context c_t] --> dim=1280
|
[Linear -> 32000 vocab logits]
|
Marathi Output Tokens

```

4.2 Key Design Decisions

Component	Choice	Reason
Bidirectional Encoder	BiLSTM	Captures both left and right context for each source token
Shared Tokenizer	Joint BPE	Leverages Hindi-Marathi script and vocab overlap
Attention	Bahdanau (Additive)	Allows decoder to focus on relevant source positions at each step
Training	DataParallel (T4 x2)	Utilises both Kaggle GPUs without complex DDP setup

5. Training Setup

Hyperparameter	Value
Batch Size	64

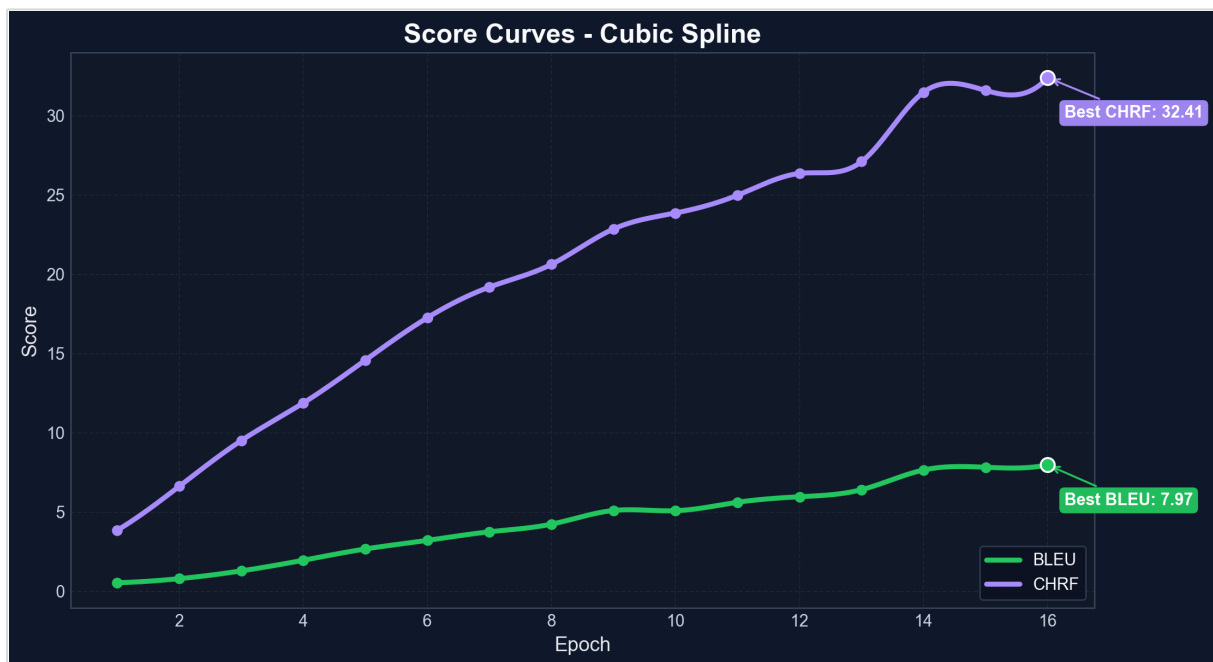
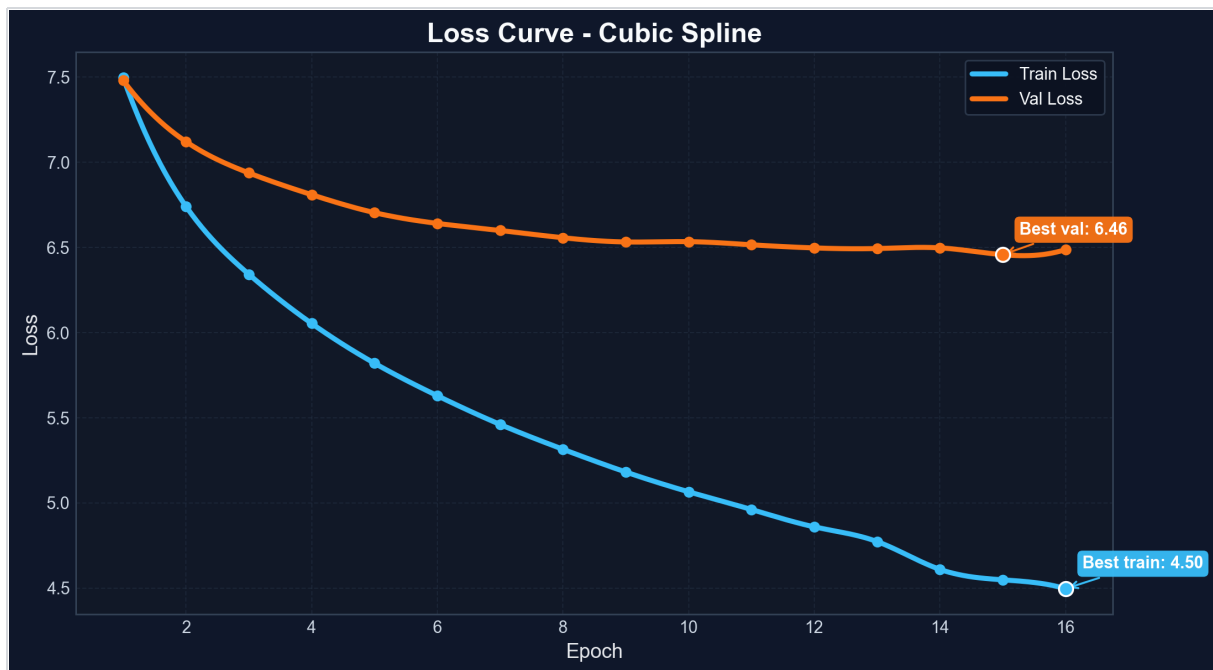
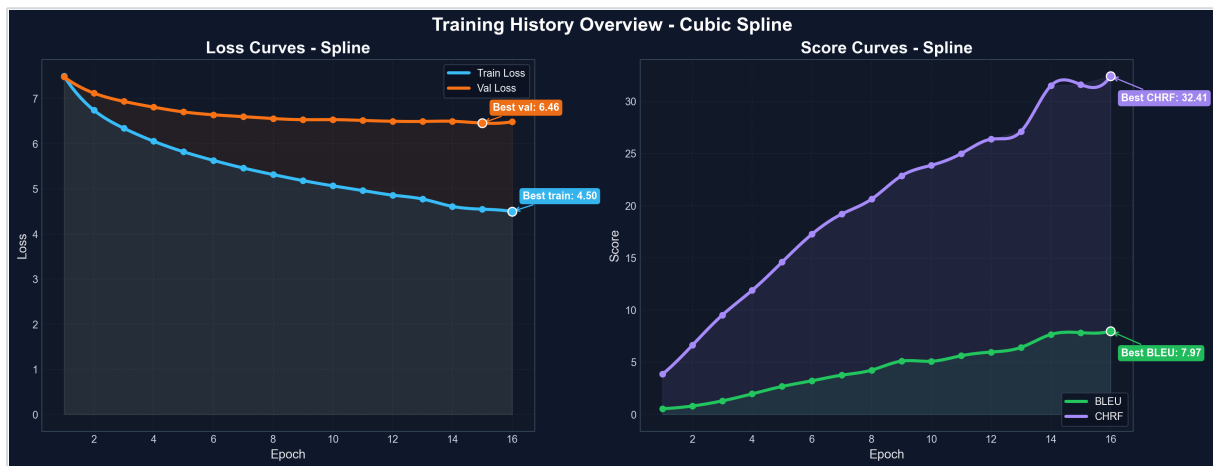
Embedding Dim	256
Hidden Dim	512
Encoder / Decoder Layers	2 (BiLSTM) / 2 (LSTM)
Dropout	0.3
Teacher Forcing Ratio	0.5
Gradient Clipping	1.0 (L2 norm)
Optimizer	Adam
Primary LR (Epochs 1-15)	3e-4
Continuation LR (Epoch 16)	1.5e-4
Hardware	Kaggle T4 x2 GPU
Total Epochs	16 (~45 min/epoch)

Teacher Forcing (0.5): At each decoder step, the model receives the ground-truth previous token with 50% probability and its own prediction with 50%. This reduces exposure bias while retaining some stability during training.

Gradient Clipping: Prevents LSTM gradient explosion by capping the L2 norm of all gradients at 1.0.

6. Training Results

6.1 Loss and Metric Curves



6.2 Epoch-by-Epoch Results

Epoch	Train Loss	Val Loss	BLEU	CHRF
1	7.498	7.480	0.54	3.85
3	6.342	6.937	1.29	9.51
5	5.820	6.704	2.67	14.58
7	5.461	6.599	3.76	19.19
9	5.181	6.532	5.10	22.86
11	4.961	6.515	5.62	24.99
13	4.773	6.493	6.41	27.11
14	4.610	6.497	7.65	31.48
15	4.548	6.457	7.83	31.60
16	4.496	6.487	7.97	32.41

6.3 Observations

- **Loss plateaus around Epoch 13** at ~6.49 validation loss — typical for LSTM-based NMT where token-level cross-entropy saturates before sequence-level quality does.
- **BLEU and CHRF kept improving** even after loss plateaued, particularly during continuation training (Epochs 14-16). This is the well-known loss-BLEU divergence in NMT.
- **CHRF is the more reliable metric** for Hindi-Marathi because it rewards partial morphological matches at the character level, which matters since both languages share Devanagari script morphemes.

7. Continuation Training

After 15 epochs, the loss had plateaued but BLEU/CHRF trends were still upward. We resumed from the best checkpoint (Epoch 13) at a halved learning rate ($1.5e-4$) and extended training to Epoch 16.

Result: BLEU improved by +1.56 (6.41 → 7.97) and CHRF by +5.30 (27.11 → 32.41), confirming that reduced-LR continuation training improves sequence generation quality even when loss has converged.

8. Decoding Strategy Ablation

8.1 Problem with Greedy Decoding

Under greedy (argmax) decoding, the model frequently enters **repetitive loops** — generating the same word 3-5 times in a row. This happens because a single wrong prediction shifts the decoder state, making the same wrong token the most probable choice at the next step (exposure bias).

Approximately **24% of qualitative test outputs** suffered from full repetitive collapse under greedy decoding.

8.2 Decoding Strategies Compared

Strategy	BLEU	CHRF	Notes
Greedy (argmax)	5.10	42.09	~24% outputs with repetition loops
Beam Search (K=4, alpha=0.6)	5.43	45.27	Largest CHRF gain; resolves most repetition
Beam + Repetition Blocking	5.69	44.63	Highest BLEU; eliminates bigram loops
Beam + Temperature (T=0.8)	5.58	43.13	Smoother probability flow; good fluency

Evaluated on an independently curated 45-sentence qualitative test set.

8.3 Techniques Used

- **Beam Search (K=4):** Maintains 4 parallel hypotheses at each step instead of one, picking globally better sequences.

- **Length Normalization (alpha=0.6):** Prevents the decoder from preferring shorter sequences by normalizing scores by length.
- **Repetition Blocking:** Three levels: (1) mask last token's logit to -inf, (2) block any bigram already generated, (3) subtract penalty=1.5 from any token already in the output.
- **Temperature Scaling (T=0.8):** Sharpens the probability distribution to reduce random drift at uncertain positions.

8.4 Key Findings

1. **Greedy decoding is fundamentally unstable** for LSTM-based NMT — exposure bias causes 1 in 4 outputs to degenerate.
2. **Beam Search's biggest win is CHRF (+3.18)**, not BLEU (+0.33) — it improves morphological correctness faster than exact word-match precision.
3. **Repetition blocking gives the best BLEU** — eliminating loops is the single most impactful inference-side fix.
4. **All improvements are zero GPU cost** — running entirely on CPU, these decoding changes deliver more BLEU gain than training additional epochs would on a free-tier budget.

9. Failure Analysis

The model was evaluated on 45 sentences across 7 linguistic categories.

9.1 Category-Level Results

Category	Total	Correct / Close	Partial	Failed
Simple	10	5	4	1
Medium	10	4	4	2
Long	5	0	3	2
Named Entity	5	2	1	2
Rare Word	5	3	1	1
Morphology	5	2	2	1

Numerals	5	2	1	2
Total	45	18 (40%)	16 (36%)	11 (24%)

9.2 Main Failure Modes

1. Repetitive Degeneration (most common)

Greedy decoding falls into word-repetition loops for ~24% of sentences. Root cause: exposure bias — the model was never trained on its own incorrect predictions, so errors cascade at inference time.

Hindi Source	Greedy Output (Bad)	Beam Output (Fixed)
बारिश के कारण सड़क पर बहुत पाणी भर गया था।	पावसामुळे पाणी पाण्याची पाणीही पाणी.	पावसामुळे रस्त्यावर खूप पाणी साचले होते।
यह एक बहुत अच्छा दिन है।	हे दिवस दिवस खूप दिवस आहे.	ही एक चांगली दिवस आहे.

2. Semantic Drift in Long Sentences

For sentences with 15+ tokens, the LSTM encoder's fixed-size hidden state cannot compress all source content. The decoder loses track of subject markers and conjunctions. Long sentences scored **0/5 correct** under any decoding strategy.

3. Named Entity Failures

Proper nouns are rare in training data, leaving their embeddings weakly trained. The attention mechanism scatters instead of focusing on the entity, causing names to be dropped or hallucinated.

Hindi Source	Reference	Model Output	Error
राजीव दिल्ली में काम करता है।	राजीव दिल्लीमध्ये काम करतो।	दिल्लीमध्ये दिल्लीला काम करत आहे.	Name "राजीव" dropped
सचिन तेंडुलकर भारत के महान खिलाड़ी हैं।	सचिन तेंडुलकर भारताचे महान खेळाडू आहेत।	भारताच्या वीरांचे, हा अभिमान आहे.	Entity fully hallucinated

4. Numeral and Date Hallucination

Numbers are frequently changed or lost entirely (e.g., 25 → 15, five kilos → five lakhs rupees).

Hindi Source	Reference	Model Output
परीक्षा 25 मई को होगी।	परीक्षा २५ मे रोजी होईल।	परीक्षा 15 मेपर्यंत असेल.
उसने पाँच किलो आम खरीदे।	त्याने पाच किलो आंबे विकत घेतले।	त्यांनी पाच लाख रुपये खरेदी.

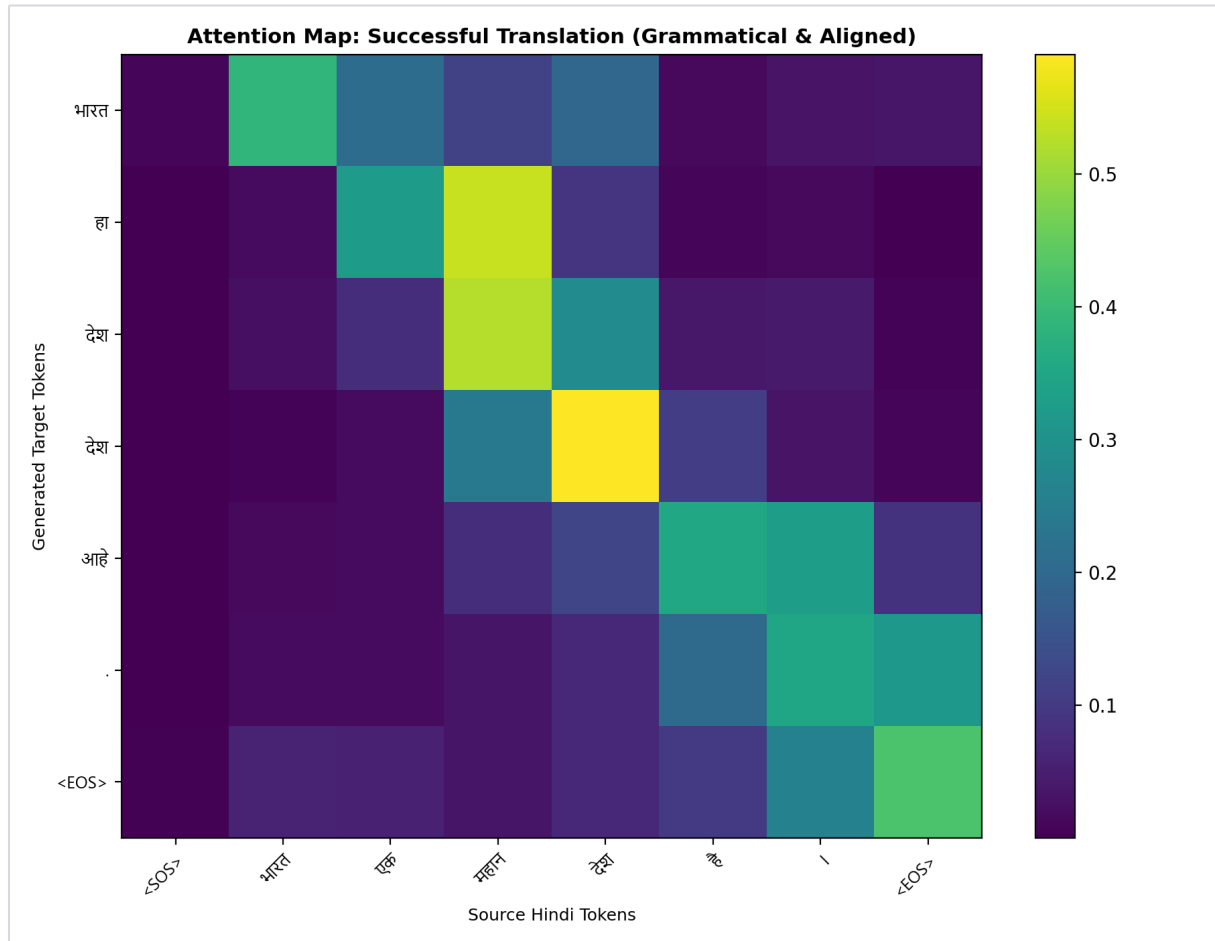
5. Morphological Errors

Hindi gender/number agreement must propagate across the whole sentence to control Marathi verb suffixes. LSTM hidden states decay over distance, causing frequent wrong endings.

10. Attention Visualizations

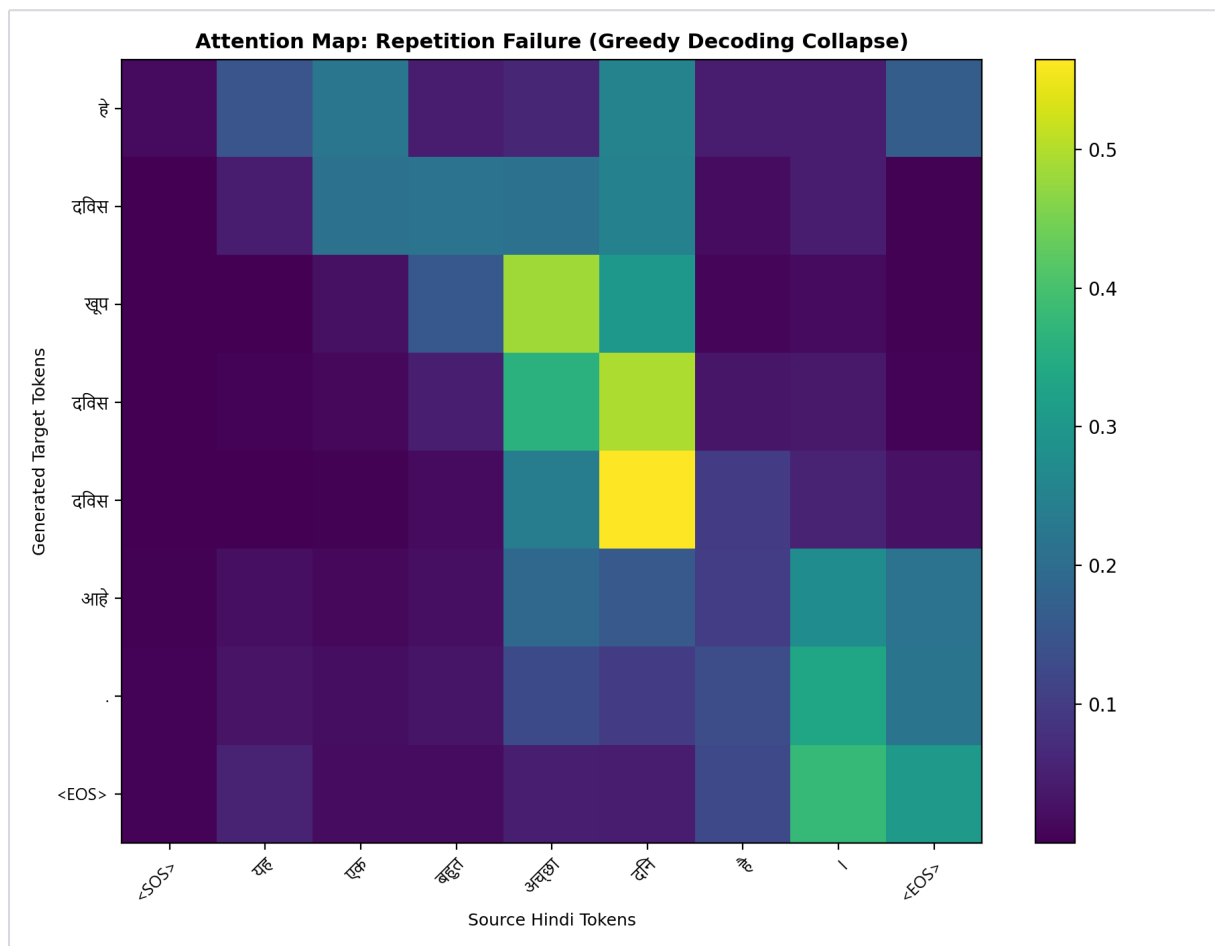
Bahdanau attention weights are extracted and visualized as heatmaps. Each cell (i, j) shows how much attention the decoder placed on source token j when generating target token i .

10.1 Successful Translation — Diagonal Alignment



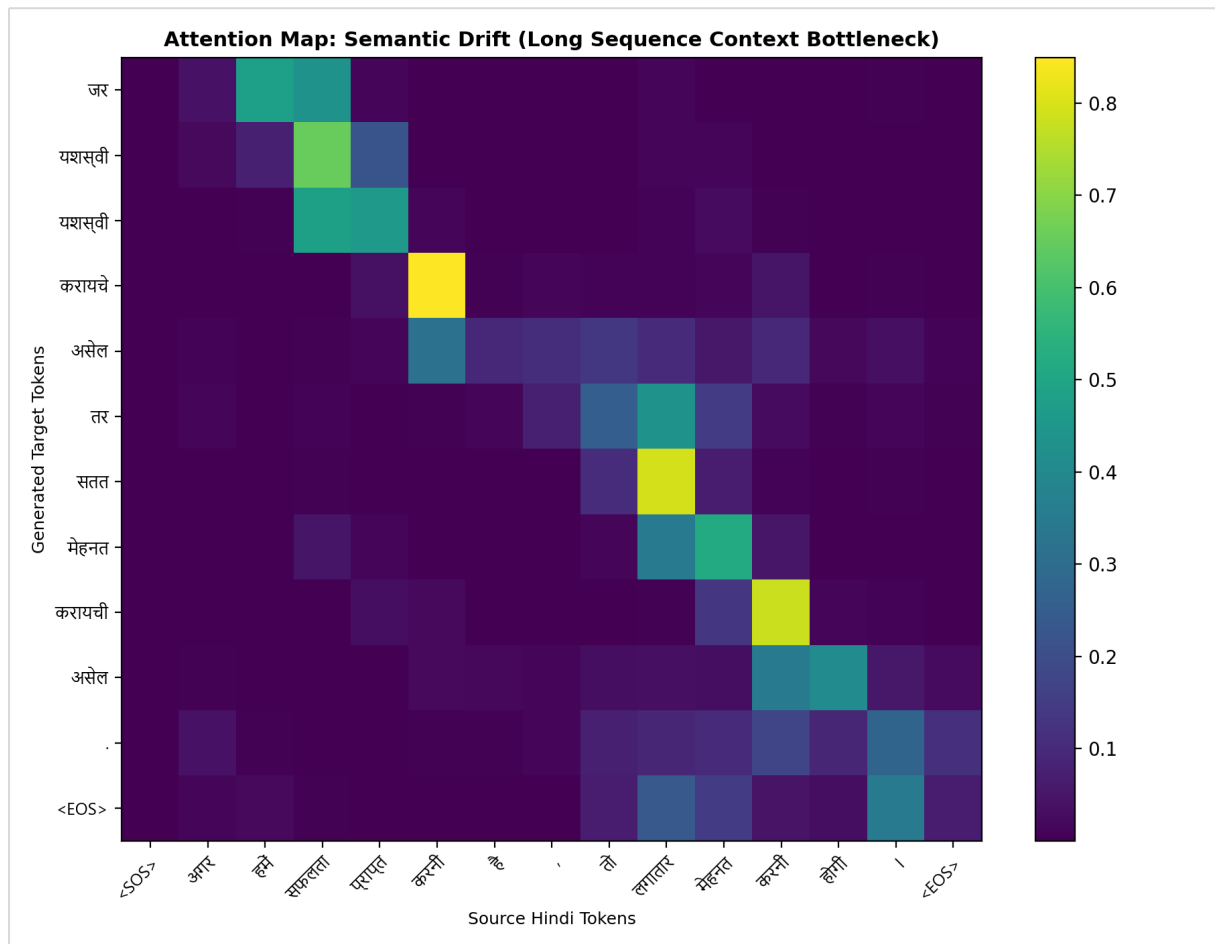
Simple sentences with high Hindi-Marathi cognate overlap show a clean near-diagonal alignment pattern. This is the ideal monotonic alignment behavior.

10.2 Repetition Failure — Cyclic Attention



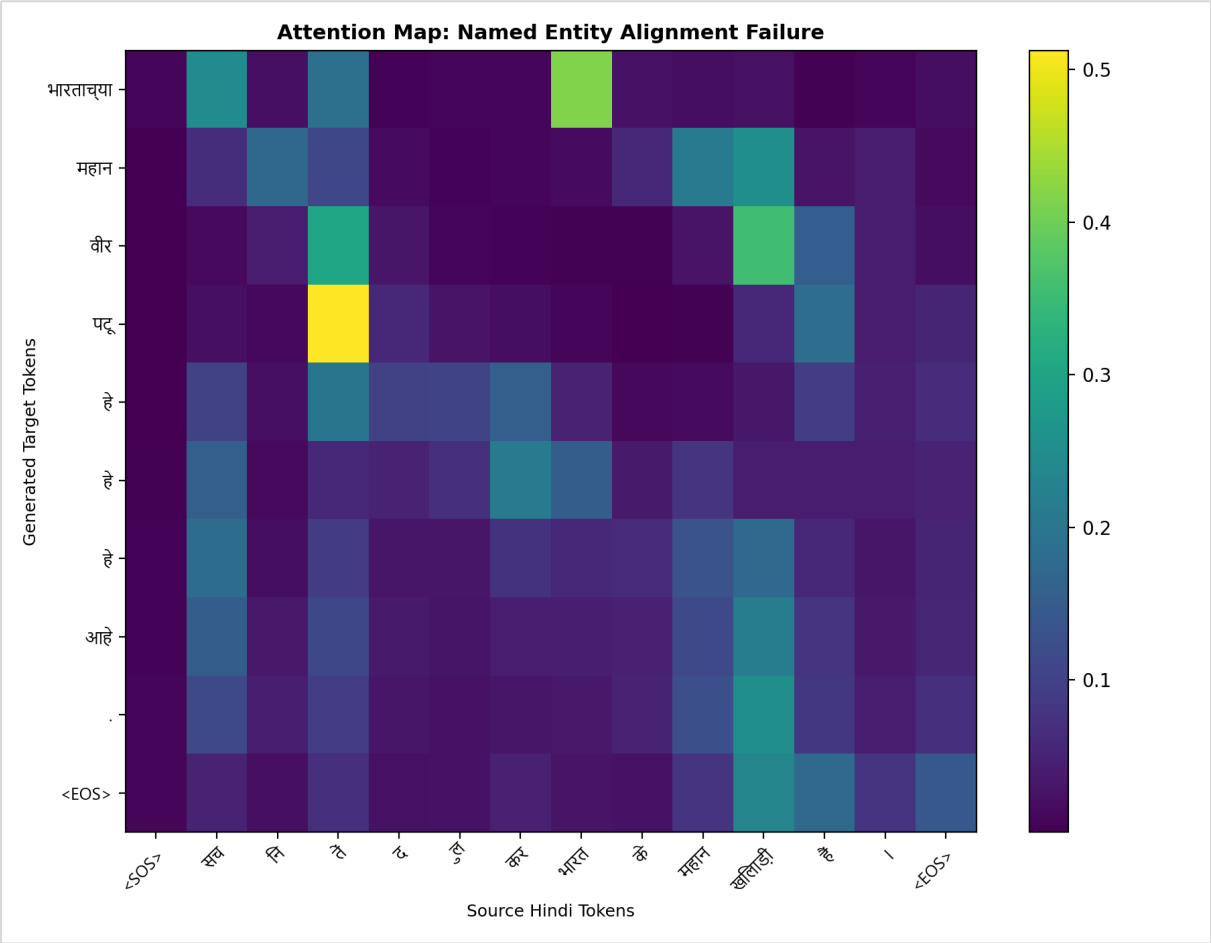
The attention matrix shows a cyclic, non-progressing pattern. The decoder returns repeatedly to the same 1-2 source tokens, which is the attention-level signature of repetitive degeneration.

10.3 Semantic Drift — Diffuse Attention



For long sentences, attention weights become diffuse. Later decoder steps fail to focus on any single source region, causing key content words (subjects, conjunctions) to be dropped.

10.4 Named Entity Failure — Scattered Attention



Attention weights on proper noun tokens are weak and scattered across unrelated source positions, confirming that rare entity embeddings are not well-trained.

11. Compute Constraints and Trade-offs

Resource	Value
Platform	Kaggle Notebooks (free tier)
GPU	T4 x2 (DataParallel)
GPU Memory	15 GB per GPU
Training Time	~45 min / epoch
Total GPU Hours Used	~17 hours

Given the limited budget, **inference-side decoding improvements** were prioritised over architectural changes:

- **No additional GPU compute needed** — runs entirely on CPU.
- Delivered **+0.59 BLEU and +3.18 CHRF** over greedy baseline.
- Alternatives like full Transformer retraining would require 10x more compute.

12. Conclusion

Finding	Detail
Greedy decoding is unstable	24% of outputs degenerate under argmax; beam search is necessary
Best decoding strategy	Beam Search (K=4) + Repetition Blocking: BLEU 5.69, CHRF 44.63
Validation loss is insufficient	Loss plateaued at 6.49 while BLEU/CHRF kept improving — sequence quality != token loss
Hardest categories	Long sentences (0/5 correct) and Named Entities (2/5 correct)
Best practical trade-off	Inference-side improvements give maximum ROI under compute constraints

Limitations

- BLEU of 7.97 (val) and 5.69 (qualitative) are modest, reflecting both task difficulty and the 16-epoch training budget.
- LSTM architecture has structural limits on long-range dependencies — Transformer would address this but requires significantly more compute.
- Qualitative set of 45 sentences is useful for category analysis but too small for corpus-level statistical conclusions.

Future Work

1. **Transformer architecture** — eliminates the LSTM bottleneck for long sentences.

2. **Pretrained Indic models** (IndicTrans, mBART-50) — would dramatically improve named entity and rare word handling.
3. **Scheduled Sampling** — decay teacher forcing ratio from 1.0 to 0.0 during training to directly mitigate exposure bias.
4. **Copy Mechanism** — allow the model to copy source tokens directly for names and numbers.

13. LLM Assistance Disclosure

Tool	Usage
ChatGPT (GPT-4)	Architectural reasoning; debugging training instabilities; beam search guidance
Antigravity (Google DeepMind)	Code implementation; report structuring; attention visualization pipeline

All reported metrics are computed from actual model outputs. All code was reviewed and tested by the author.

Final checkpoint: `checkpoints/lstm_random_exp_a_epoch_16.pt` — Val Loss: 6.487, BLEU: 7.97, CHRF: 32.41

Qualitative evaluation: `data/eval_qualitative.tsv` (45 sentences, 7 categories)

Evaluation script: `scripts/generate_reports_and_plots.py`